Probability Rules & Conditional Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

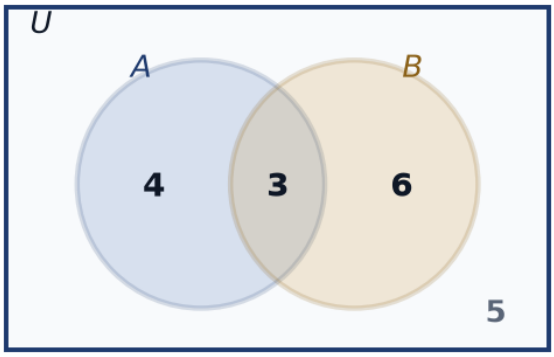
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic probability rules

A fair six-sided die is rolled. Find the probability the roll is even, and the probability the roll is greater than 4.

2 [Foundational] • Target: Basic probability rules

For events A and B, $P(A) = 0.4$, $P(B) = 0.3$, and $P(A \cup B) = 0.6$. Find $P(A \cap B)$ using the addition rule.



3 [Foundational] • Target: Mutually exclusive and independent events

Explain the difference between mutually exclusive events and independent events, and give a brief example of each.

4 [Foundational] • Target: Mutually exclusive and independent events

Two events A and B satisfy $P(A) = 0.5$ and $P(B) = 0.5$, and they are mutually exclusive. Explain why A and B cannot also be independent (unless one has probability 0).

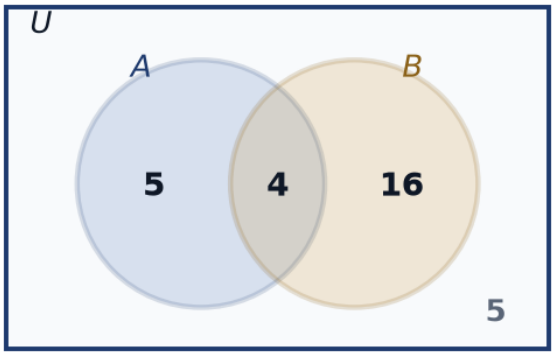
5 [Foundational] • Target: Conditional probability

In a class, the probability a student likes math is 0.6, and the probability a student likes both math and science is 0.42. Find the probability a student likes science given that they like math.

© ClipSAT — clipsat.org | All Rights Reserved

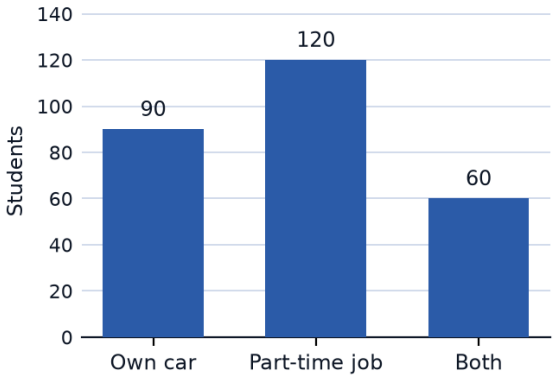
6 [Foundational] • Target: Conditional probability

Use the multiplication rule to find $P(A \cap B)$ if $P(A) = 0.5$ and $P(B \mid A) = 0.4$.



7 [Foundational] • Target: Two-way tables and conditional probability

A survey of 200 students finds 90 own a car, 120 have a part-time job, and 60 both own a car and have a job. Find the probability a randomly selected student has a job, given that they own a car.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Two-way tables and conditional probability

Using the same survey data, determine whether owning a car and having a job are independent events, showing your reasoning.

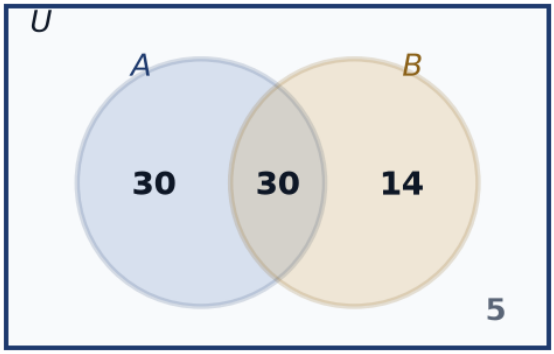
9 [Intermediate] • Target: Tree diagrams and sequential events

A factory has two machines: Machine A produces 60% of items with a 2% defect rate, and Machine B produces 40% of items with a 5% defect rate. Find the probability a randomly selected item is defective, using the law of total probability.

© ClipSAT — clipsat.org | All Rights Reserved

10 [Intermediate] • Target: Applying probability rules to real contexts

A weather forecaster states there is a 30% chance of rain on Saturday and a 30% chance of rain on Sunday, and treats the two days as independent. Find the probability it rains on both days, and the probability it rains on at least one of the two days.



11 [Intermediate] • Target: The complement rule

A box of 50 light bulbs contains 6 defective bulbs. If one bulb is selected at random, find the probability it is NOT defective, using the complement rule.

12 [Intermediate] • Target: The complement rule

A student estimates there is a 65% chance of rain at least once during a 3-day trip, assuming each day is independent with the same daily rain probability p. Find p, to three decimal places, using $P(\text{no rain all 3 days}) = (1 - p)^3$.

13 [Intermediate] • Target: Combining independence and conditional probability

Two lie detector tests are given to a suspect, each correctly detecting deception 90% of the time (independently of each other). If the suspect is actually lying, find the probability both tests correctly detect it.

14 [Intermediate] • Target: Combining independence and conditional probability

A bag contains 5 red marbles and 3 blue marbles. Two marbles are drawn without replacement. Find the probability both are red.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

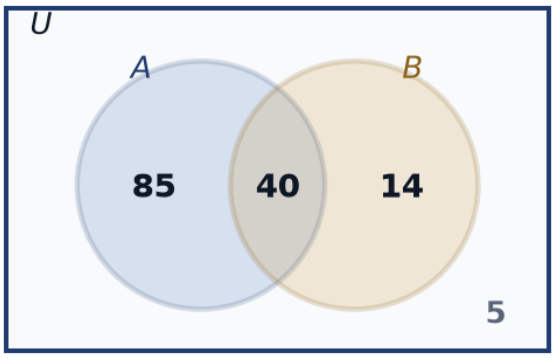
© ClipSAT — clipsat.org | All Rights Reserved

15 [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

At a certain hospital, 40% of patients admitted have condition A, and 25% have condition B. Additionally, 10% of all patients have both conditions A and B. Draw on your understanding of the addition rule to find the probability a randomly selected patient has condition A or condition B (or both), and separately explain why simply adding $P(A) + P(B)$ without adjustment would overcount certain patients.

16 [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 1.01 Measures of Center & Spread

An airline reports that 85% of its flights depart on time. Among on-time flights, 95% also arrive on time, while among delayed departures, only 40% arrive on time. Using a tree-diagram approach and the law of total probability, find the overall probability that a randomly selected flight arrives on time.



17 [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A quality inspector at a factory knows that 3% of all items produced are defective. The inspection process correctly identifies a defective item 98% of the time, but also incorrectly flags a non-defective item as defective 2% of the time (a false positive). If an item is flagged as defective by the inspection process, use the given information and the multiplication rule to find the probability that the item actually IS defective, showing every step of your reasoning (this requires combining the law of total probability with conditional probability, similar to Bayes' theorem).

18 [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A survey of 500 adults finds that 280 exercise regularly, 220 eat a balanced diet, and 150 do both. Using this information, first find the probability a randomly selected adult exercises regularly or eats a balanced diet (or both), and then find the probability an adult eats a balanced diet given that they exercise regularly, showing every step of your work clearly.

19 [Challenging] • Target: The complement rule • Cross-Lesson: 2.01 Sampling Methods & Bias

Events A and B satisfy $P(A) = 0.40$, $P(B) = 0.50$, and $P(A \cup B) = 0.70$. (a) Find $P(A \cap B)$. (b) Determine whether A and B are independent. (c) Determine whether A and B are mutually exclusive.

© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A national health organization is investigating the relationship between daily screen time (in hours) and the number of hours of sleep reported by adults. They collect data from a sample of 1200 adults. The data is summarized in a two-way table, where the row totals represent categories of daily screen time and the column totals represent categories of reported sleep hours. The organization is particularly interested in the cells where adults report less than 4 hours of screen time and get 7 or more hours of sleep. If the overall row total for 'less than 4 hours of screen time' is 250, and the overall column total for '7 or more hours of sleep' is 300, what is the expected count for the cell representing adults with less than 4 hours of screen time who get 7 or more hours of sleep, assuming independence between screen time and sleep hours?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic probability rules

For events A and B, $P(A) = 0.30$, $P(B) = 0.70$, and $P(A \cap B) = 0.05$. Find $P(A \cup B)$ using the addition rule.

a. 0.90

b. 1.05

c. 0.95

d. 1.00

22 [Foundational] • Target: Mutually exclusive and independent events

For events A and B, $P(A) = 0.35$, $P(B) = 0.60$, and $P(A \cap B) = 0.25$. Find $P(A \cup B)$ using the addition rule.

a. 0.70

b. 0.95

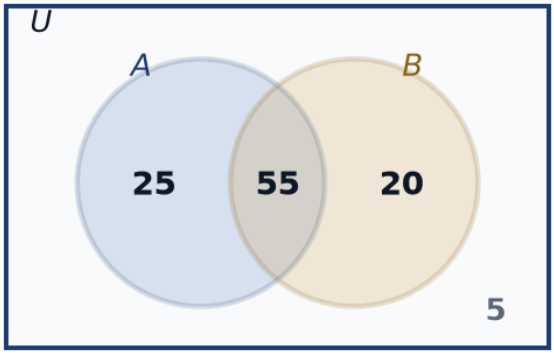
c. 0.75

d. 0.65

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Conditional probability

For events A and B, $P(A) = 0.25$, $P(B) = 0.55$, and $P(A \cap B) = 0.20$. Find $P(A \cup B)$ using the addition rule.



a. 0.55

b. 0.80

c. 0.65

d. 0.60

24 [Foundational] • Target: Two-way tables and conditional probability

For events A and B, $P(A) = 0.35$, $P(B) = 0.50$, and $P(A \cap B) = 0.10$. Find $P(A \cup B)$ using the addition rule.

a. 0.75

b. 0.80

c. 0.70

d. 0.85

25 [Foundational] • Target: Tree diagrams and sequential events

For events A and B, $P(A) = 0.35$, $P(B) = 0.55$, and $P(A \cap B) = 0.30$. Find $P(A \cup B)$ using the addition rule.

a. 0.65

b. 0.90

c. 0.55

d. 0.60

26 [Foundational] • Target: Applying probability rules to real contexts

For events A and B, $P(A) = 0.60$, $P(B) = 0.65$, and $P(A \cap B) = 0.25$. Find $P(A \cup B)$ using the addition rule.

a. 0.95

b. 1.00

c. 1.05

d. 1.25

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: The complement rule

For events A and B, $P(A) = 0.55$, $P(B) = 0.45$, and $P(A \cap B) = 0.45$. Find $P(A \cup B)$ using the addition rule.

a. 0.50

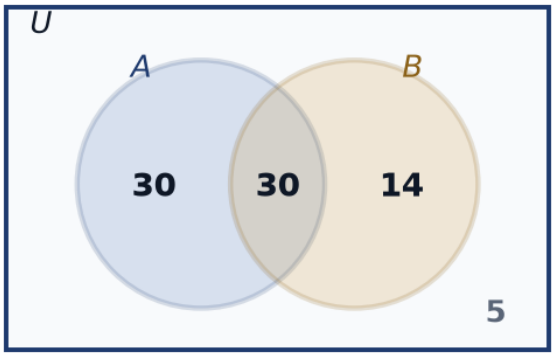
b. 0.55

c. 0.60

d. 1.00

28 [Foundational] • Target: Combining independence and conditional probability

For events A and B, $P(A) = 0.30$ and $P(A \cap B) = 0.30$. Find $P(B \mid A)$.



a. 1.10

b. 1.00

c. 0.90

d. 0.00

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: General Addition Rule & Mutually Exclusive Events

For events A and B, $P(A) = 0.30$ and $P(A \cap B) = 0.10$. Find $P(B \mid A)$.

a. 0.43

b. 0.20

c. 0.33

d. 0.23

30 [Intermediate] • Target: Conditional Probability Formula

For events A and B, $P(A) = 0.80$ and $P(A \cap B) = 0.40$. Find $P(B \mid A)$.

a. 0.60

b. 0.50

c. 0.40

d. 0.55

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Multiplication Rule and Tree Diagrams

For events A and B, $P(A) = 0.65$ and $P(A \cap B) = 0.60$. Find $P(B \mid A)$.

a. 0.92

b. 1.02

c. 0.82

d. 0.05

32 [Intermediate] • Target: Bayes' Theorem in Context

For events A and B, $P(A) = 0.70$ and $P(A \cap B) = 0.60$. Find $P(B \mid A)$.

a. 0.10

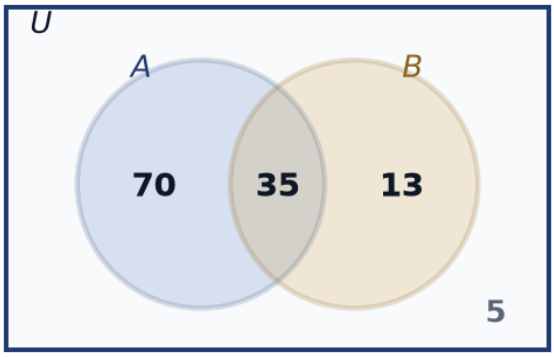
b. 0.86

c. 0.76

d. 0.96

33 [Intermediate] • Target: Basic probability rules

For events A and B, $P(A) = 0.70$ and $P(A \cap B) = 0.35$. Find $P(B \mid A)$.



a. 0.50

b. 0.35

c. 0.60

d. 0.40

34 [Intermediate] • Target: Mutually exclusive and independent events

For events A and B, $P(A) = 0.35$ and $P(A \cap B) = 0.10$. Find $P(B \mid A)$.

a. 0.29

b. 0.39

c. 0.25

d. 0.19

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Conditional probability

A bag contains 7 red marbles and 6 blue marbles. Two marbles are drawn without replacement. Find the probability both are blue.

a. $\frac{19}{65}$

b. $\frac{6}{65}$

c. $\frac{6}{13}$

d. $\frac{5}{26}$

36 [Intermediate] • Target: Two-way tables and conditional probability

A bag contains 4 red marbles and 8 blue marbles. Two marbles are drawn without replacement. Find the probability both are blue.

a. $\frac{107}{330}$

b. $\frac{14}{33}$

c. $\frac{2}{3}$

d. $\frac{173}{330}$

37 [Intermediate] • Target: Tree diagrams and sequential events

A factory has two machines: Machine A produces 80% of items with a 10% defect rate, and Machine B produces 20% of items with a 3% defect rate. Letting D denote the event that an item is defective, find $P(D)$ for a randomly selected item, using the law of total probability.

a. 8.60%

b. 4.40%

c. 8.10%

d. 9.10%

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applying probability rules to real contexts • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A factory has two machines: Machine A produces 50% of items with a 9% defect rate, and Machine B produces 50% of items with a 2% defect rate. Letting D denote the event that an item is defective, find $P(D)$ for a randomly selected item, using the law of total probability.

a. 6.00%

b. 5.50%

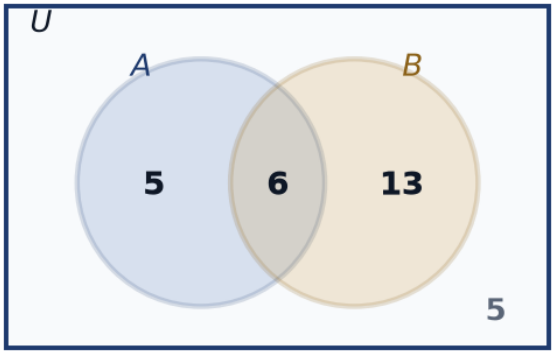
c. 6.50%

d. 5.00%

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: The complement rule • Cross-Lesson: 2.01 Sampling Methods & Bias

A quality inspector knows 5% of all items produced are defective (event D). The inspection process correctly flags a defective item 93% of the time, but also flags a non-defective item as defective 6% of the time (a false positive). If an item is flagged as defective (event F), find $P(D \mid F)$, to one decimal place.



a. 93.0%

b. 47.9%

c. 44.9%

d. 41.9%

40 [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A quality inspector knows 3% of all items produced are defective (event D). The inspection process correctly flags a defective item 97% of the time, but also flags a non-defective item as defective 6% of the time (a false positive). If an item is flagged as defective (event F), find $P(D \mid F)$, to one decimal place.

a. 30.3%

b. 97.0%

c. 33.3%

d. 36.3%

41 [Challenging] • Target: General Addition Rule & Mutually Exclusive Events • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

If $P(A) = 0.40$, $P(B) = 0.35$, and $P(A \cap B) = 0.15$, what is $P(A \cup B)$?

a. 0.60

b. 0.75

c. 0.90

d. 0.25

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: General Addition Rule & Mutually Exclusive Events • Cross-Lesson: 3.03 The Normal Distribution

Can two events with non-zero probabilities be both mutually exclusive and independent?

a. No, if they are mutually exclusive, the occurrence of one prevents the other, meaning they are dependent

b. Yes, mutually exclusive and independent are synonyms in probability

c. Only if the sum of their probabilities equals 1

d. Yes, as long as their intersection is zero

43 [Challenging] • Target: Conditional Probability Formula • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

Which statement mathematically proves that two events A and B are independent?

a. $P(A \mid B) = P(A)$

b. $P(A \cup B) = P(A) + P(B)$

c. $P(A \cap B) = 0$

d. $P(A) = P(B)$

44 [Challenging] • Target: Multiplication Rule and Tree Diagrams • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

If two events C and D are independent with $P(C) = 0.60$ and $P(D) = 0.70$, what is $P(C \cap D)$?

a. 0.42

b. 1.30

c. 0.10

d. 0.65

45 [Challenging] • Target: Bayes' Theorem in Context • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

Continuing the factory problem: if an item is found to be defective, what is the probability it came from Machine A?

a. $\frac{0.012}{0.032} = 0.375$

b. $\frac{0.60}{0.032}$

c. 0.60

d. 0.020

© ClipSAT — clipsat.org | All Rights Reserved